EXPERIMENT - 22

22. Implement the Closing technique as a Morphological operation to dilate the foreground regions based on Open CV.

Aim

To implement the Closing technique as a morphological operation using OpenCV in Python to fill small holes and close gaps in foreground regions.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Resize the image to fit the screen.
4. Convert the image into grayscale.
5. Create a kernel using np.ones().
6. Apply the Closing operation using cv2.morphologyEx() with cv2.MORPH_CLOSE.
7. Display the original image and the closing output image.

Program

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

if img is None:
    print("Image not found")
else:
    img = cv2.resize(img, (500, 350))
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

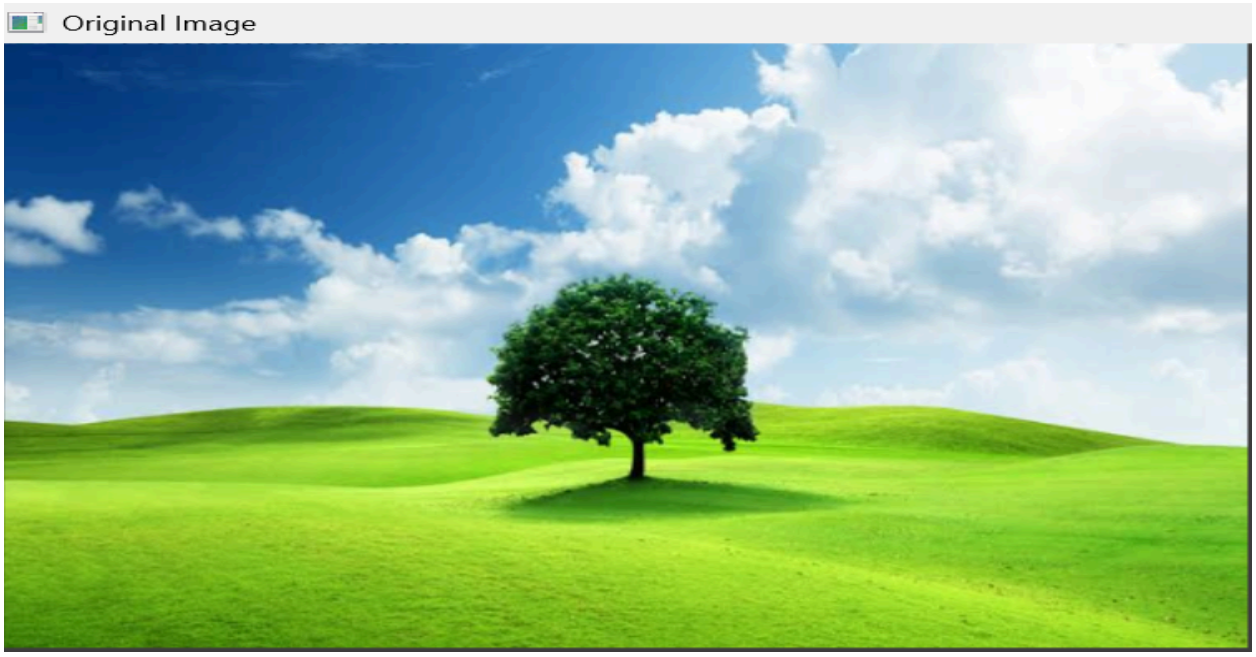
    kernel = np.ones((5, 5), np.uint8)

    closing = cv2.morphologyEx(gray, cv2.MORPH_CLOSE, kernel)

    cv2.imshow("Original Image", img)
    cv2.imshow("Closing Output", closing)
```

```
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input Image



Output Image



Result

The Closing Morphological Operation was successfully implemented using OpenCV in Python.